FINAL DATA ANALYTICS PROJECT

Supermarket Sales Data Analytics

A complete case study covering data preparation, exploratory analysis and interactive visualization

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Organization: SWYNEX Technologies
Internship: Data Analyst Internship
Tools: Python, Pandas, Matplotlib, Seaborn, Google Looker Studio, GitHub

Project report | November 2026

1. Executive Summary

This report brings together the work completed across the SWYNEX Data Analyst Internship tasks using the Supermarket Sales dataset. The workflow covers data cleaning and preparation, exploratory data analysis (EDA), and the development of an interactive dashboard in Google Looker Studio.

The dataset contains 1,000 transaction records. The analysis reported total revenue of 322,966.75, total quantity sold of 5,510, an average transaction value of 322.97, and an average customer rating of 6.97. No missing values or duplicate rows were identified in the reviewed dataset.

The dashboard provides an at-a-glance view of revenue, quantity, customer rating and transaction count, alongside product-line, branch, payment-method and weekday analyses. Interactive filters allow users to explore the results by branch, product line, payment method and date.

The report is intended to document the completed analysis and communicate findings clearly. It does not claim causal relationships or make predictions from the descriptive analysis.

Project at a glance

Item	Details
Dataset	Supermarket Sales
Records	1,000
Original columns	17
Total revenue	322,966.75
Total quantity	5,510
Average transaction value	322.97
Average customer rating	6.97
Detected outliers	9 (IQR method)

2. Problem Statement and Objectives

Retail businesses generate transaction data that can be used to understand sales performance, product demand, customer purchasing patterns and differences between branches. However, raw transaction records are difficult to interpret without preparation and summarization.

Problem statement: Analyze supermarket transaction data to summarize sales performance and identify descriptive patterns across product lines, branches, payment methods and weekdays. Present the findings in a form that is easy for business users to explore.

Objectives

• Inspect and prepare the supermarket sales dataset for analysis.

• Summarize important numerical measures and check data quality.

• Explore sales patterns across product lines, branches, payment methods and weekdays.

• Identify potential outliers using the Interquartile Range (IQR) method.

• Build an interactive dashboard with KPIs, charts and filters.

• Communicate findings and limitations in a documented case study.

Scope

The project is a descriptive analysis of the supplied supermarket sales dataset. It focuses on the available transaction fields and the analysis already completed in Tasks 1–3. It does not include customer-level segmentation, forecasting, causal inference or recommendations based on external market data.

3. Dataset Description

The project uses the Supermarket Sales CSV dataset. The source dataset was described in the internship work as containing 1,000 records and 17 original columns. A derived weekday field was added during the EDA/dashboard preparation, which is why some analysis outputs may show an additional column.

The transaction data supports analysis of sales amount, quantity, product line, branch, payment method, date and customer rating. The cleaned CSV is included in the Task 4 GitHub project folder.

Data quality review

Check	Reported result	Interpretation
Record count	1,000	Transactions included in analysis
Missing values	0	No missing values reported
Duplicate rows	0	No duplicate rows reported
Original columns	17	Before adding derived weekday field
Derived field	Day of week	Created from transaction date for weekday analysis

The data-quality checks provide a baseline for the subsequent analysis. A zero count in these checks does not guarantee that every value is semantically correct; validation of business rules and source-system accuracy would require additional context.

4. Data Cleaning and Preparation

Data cleaning and preparation were performed using Python and the Pandas library in Google Colab. The purpose was to inspect the dataset, check for common quality issues and create a clean file for further analysis.

Preparation workflow

• Load the source CSV into a Pandas DataFrame.

• Inspect the dataset structure, columns and data types.

• Check for missing values.

• Check for duplicate records.

• Review the data and prepare the cleaned dataset for EDA.

• Save the prepared data as cleaned_supermarket_sales.csv.

Data quality results

Metric	Result
Missing values	0
Duplicate rows	0
Records analyzed	1,000
Cleaned output	cleaned_supermarket_sales.csv

The cleaning notebook (data_cleaning.ipynb) and cleaned CSV are included in the Task 4 repository so the preparation workflow and output can be reviewed.

Reproducibility note: the exact code cells and transformation details should be consulted in the included notebook. This report summarizes the documented checks and does not invent additional cleaning operations.

5. Exploratory Data Analysis

Exploratory Data Analysis was used to summarize the transaction data and examine patterns across key business dimensions. The work included descriptive statistics, visualizations and outlier detection.

Summary statistics

Measure	Result
Total revenue	322,966.75
Average sales / transaction value	322.97
Maximum transaction value	1,042.65
Minimum transaction value	10.68
Average customer rating	6.97
Total quantity sold	5,510

Analysis views

• Product-line sales: compares revenue generated by the available product categories.

• Branch sales: compares total sales across supermarket branches.

• Payment distribution: examines the share of transactions by payment method.

• Sales distribution: helps inspect the spread of transaction values.

• Daily sales: compares activity across weekdays.

The five EDA chart images and EDA_analysis.ipynb are included in the repository. The dashboard and EDA notebook should be used together when reviewing the visual patterns.

6. Outlier Detection

The Interquartile Range (IQR) method was used to flag unusually high or low transaction values for review. IQR is calculated as the third quartile (Q3) minus the first quartile (Q1). The lower and upper fences are calculated as Q1 – 1.5 × IQR and Q3 + 1.5 × IQR.

IQR measure	Reported value
First quartile (Q1)	124.42
Third quartile (Q3)	471.35
Interquartile range (IQR)	346.93
Lower fence	-395.97
Upper fence	991.74
Flagged outliers	9

Nine records were flagged by the IQR rule. An outlier flag does not automatically mean a record is an error. High-value transactions may be genuine purchases. Such records should be investigated using business context before any removal or correction.

Interpretation

The IQR method is a statistical screening technique. It helps focus attention on values that are far from the middle 50% of observations, but it does not explain why those values occurred.

7. Interactive Dashboard

An interactive supermarket sales dashboard was created using Google Looker Studio. It combines KPI scorecards, charts and filters so users can explore the dataset without manually editing the source file.

Dashboard KPIs

KPI	Displayed result
Total Revenue	322,966.75 (may display as 322.97K)
Total Quantity Sold	5,510
Average Customer Rating	6.97
Total Transactions	1,000

Charts and filters

• Sales by Product Line

• Sales by Branch

• Payment Method Distribution

• Daily Sales Analysis by Weekday

• Filters for branch, product line, payment method and date range

Interactive dashboard: Open the Supermarket Sales Dashboard

The dashboard link should be tested in a signed-out or private browser window before submission to confirm that reviewers have permission to view it.

8. Key Business Insights

The completed analysis reported the following descriptive findings:

• Food and Beverages generated the highest revenue among the product lines.

• Branch C recorded the highest total sales among the branches.

• E-wallet was the most-used payment method at 34.5%, closely followed by cash at 34.4% and credit card at 31.1%.

• Saturday recorded the highest sales among weekdays; Monday was reported as the lowest.

Payment method mix

Payment method	Share
E-wallet	34.5%
Cash	34.4%
Credit card	31.1%

These findings describe the dataset analyzed. They should not be generalized to other stores, periods or customer populations without additional evidence.

Potential business uses

• Review product-line performance when planning inventory and promotions.

• Compare branch performance to identify areas for further investigation.

• Monitor payment-method usage when planning checkout options.

• Use weekday patterns as an input to staffing and operational planning, after validating the pattern over additional periods.

9. Conclusion, Limitations and Future Scope

This project demonstrates an end-to-end descriptive data analytics workflow: preparing transaction data, exploring key measures and patterns, identifying potential outliers, and presenting results through an interactive dashboard. The combined notebook, cleaned dataset, charts and dashboard provide a reviewable record of the work completed.

Limitations

• The analysis is limited to the fields and records available in the supplied dataset.

• The findings are descriptive and do not establish causes for sales differences.

• The report does not include forecasting or statistical significance testing.

• The reported patterns may not represent current supermarket operations or other locations.

Future scope

• Analyze trends over longer periods and compare monthly or seasonal performance.

• Explore customer segments and product combinations if suitable fields are available.

• Develop sales forecasting after validating data coverage and model assumptions.

• Add operational measures such as stock availability, discounts and profit margin if data becomes available.

Project resources

The Task 4 GitHub folder contains the cleaned dataset, data-cleaning notebook, EDA notebook, chart images, dashboard screenshots and this report. The interactive dashboard is linked in Section 7.

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